

POLOKWANE, South Africa

WMO: 681740

Lat: 23.8575S

Lon: 29.4517E

Elev: 4022

StdP: 12.68

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	37.3	39.3	11.8	11.7	66.5	17.5	15.4	61.4	20.6	59.2	18.0	60.1	4.1	190	0.392

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.7	91.0	62.1	88.6	62.0	86.4	62.2	69.3	79.2	68.4	77.8	67.5	76.9	8.7	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.7	114.1	72.0	66.0	111.4	71.6	64.8	106.7	71.2	36.2	79.4	35.3	77.6	34.5	77.1	73.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 18.6	16.3	14.4	DB	33.6	95.5	2.3	2.2	32.0	97.0	30.6	98.3	29.3	99.6	27.7	101.1	(4)
(5)			WB	29.5	70.9	2.0	1.2	28.1	71.8	26.9	72.5	25.8	73.1	24.4	74.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.2	72.0	72.0	69.9	65.4	60.1	55.6	54.7	59.1	65.0	68.1	69.6	71.3		
	DBStd	7.58	3.62	3.31	3.81	4.26	4.07	3.73	3.84	4.95	5.55	5.83	5.48	4.45		
	HDD50	12	0	0	0	0	1	4	6	1	0	0	0	0		
	HDD65	1141	3	1	7	46	158	283	320	193	69	35	21	5		
	CDD50	5555	681	616	618	462	314	171	151	282	450	560	589	661		
	CDD65	1207	219	197	159	58	6	0	0	9	69	130	159	201		
	CDH74	10652	1435	1308	1077	558	228	55	52	363	1154	1500	1439	1483		
	CDH80	3225	394	383	286	113	22	1	3	59	404	580	512	468		
(14) Wind	WSAvg	6.2	6.3	6.2	5.7	5.2	4.9	5.4	5.5	6.0	7.0	7.9	7.6	6.6		
(15) Precipitation	PrecAvg	19.0	3.2	2.5	2.1	1.3	0.4	0.2	0.1	0.2	0.4	1.6	3.4	3.6		
	PrecMax	37.4	10.5	13.0	6.9	5.7	2.6	2.8	1.7	3.0	3.4	5.0	8.5	7.8		
	PrecMin	10.0	0.5	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.2		
	PrecStd	4.8	2.0	2.1	1.6	1.2	0.6	0.4	0.3	0.4	0.7	1.1	1.6	1.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.0	91.7	89.8	86.4	82.8	78.7	78.6	84.5	91.5	93.6	92.8	91.6		
		MCWB	64.5	64.1	63.5	60.7	56.6	53.5	51.3	55.6	58.4	59.9	62.2	65.1		
	2%	DB	87.4	87.9	86.0	83.2	79.0	75.3	75.1	81.5	87.9	89.8	89.4	88.0		
		MCWB	65.3	64.6	63.8	60.2	55.5	52.4	51.0	54.2	57.1	60.0	61.9	64.3		
	5%	DB	84.3	84.5	83.0	80.5	76.6	72.7	72.0	78.5	84.6	86.4	86.1	85.5		
		MCWB	65.2	64.8	63.8	60.4	55.4	51.5	50.6	53.6	56.9	59.9	62.3	65.0		
	10%	DB	81.7	81.8	80.3	77.1	73.6	69.6	69.2	75.0	81.2	82.7	82.6	82.3		
		MCWB	65.5	65.1	63.6	60.4	54.9	51.2	50.2	53.1	56.4	60.2	62.6	65.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	70.6	70.2	69.7	66.1	61.6	57.6	57.0	60.0	63.1	66.5	68.5	70.2		
		MCDB	81.1	79.5	78.3	76.1	73.5	69.8	69.1	75.7	80.0	81.5	79.8	81.8		
	2%	WB	69.2	68.9	67.8	64.5	59.8	55.5	54.6	58.0	61.2	64.7	67.1	68.8		
		MCDB	78.9	78.0	76.2	74.4	71.2	67.2	67.2	73.0	78.1	79.1	78.6	79.6		
	5%	WB	68.3	67.9	66.6	63.3	58.2	54.0	52.9	56.3	59.9	63.4	66.0	67.7		
		MCDB	77.2	76.7	76.0	73.4	69.6	66.0	65.7	71.1	76.5	77.8	77.5	77.9		
	10%	WB	67.3	66.9	65.7	62.1	56.9	52.7	51.5	54.7	58.5	62.1	64.8	66.7		
		MCDB	76.0	75.8	75.2	72.1	68.3	65.1	64.6	69.2	74.7	76.4	75.7	77.1		
(35) Mean Daily Temperature Range	5% DB	MDBR	18.7	19.2	19.7	21.7	25.9	27.1	27.4	28.1	27.0	23.6	20.6	19.0		
		MCDBR	23.8	25.2	25.2	27.7	30.3	32.5	32.8	33.8	33.4	30.6	27.6	24.6		
		MCWBR	7.2	7.8	8.1	9.8	11.9	13.9	14.3	13.1	11.8	9.6	8.4	7.5		
	5% WB	MCDBR	17.9	18.2	18.7	21.2	24.1	26.1	26.8	27.3	27.3	23.9	20.9	19.8		
		MCWBR	6.4	6.7	6.8	8.3	11.0	13.0	13.5	11.9	11.1	8.9	7.9	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.376	0.374	0.365	0.341	0.306	0.291	0.288	0.341	0.397	0.416	0.385	0.383		
	taud		2.394	2.401	2.414	2.440	2.492	2.515	2.509	2.344	2.217	2.208	2.340	2.372		
	Ebn at Noon		307	303	297	290	289	288	292	285	282	287	302	305		
	Edh at Noon		41	40	37	33	29	27	28	36	45	48	43	42		
(44) All-Sky Solar Radiation	RadAvg		2075	1976	1771	1526	1432	1269	1361	1601	1846	1980	2031	2038		
	RadStd		161	180	155	126	59	81	42	77	102	136	174	148		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page